Ryan Zhenqi Zhou, Ph.D. Candidate

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EDUCATION

University at Buffalo – SUNY, USA 09/2021 - 12/2025 (expected)

Ph.D. in Geographic Information System (STEM)

- **Geographic Information System Coursework:** GeoAI, Geospatial Big Data Analytics, Cartography, Remote Sensing
- **Data Science Coursework:** Machine Learning, Deep Learning, Time Series Modeling, Database, Data Visualization

SKILLS

- **Programming/Tools:** Python, R, SQL, Git, ArcGIS Pro
- **Libraries:** NumPy, Pandas, GeoPandas, SciPy, Scikit-Learn, Keras, TensorFlow, Matplotlib, Seaborn, Bokeh
- **AI:** Regression (OLS, GWR, etc.), Tree-based Models (RF, XGBoost, GRF), Neural Networks (DNN, etc.), Clustering (K-means), Time Series Models (LSTM, Transformer, etc.), Explainable AI (SHAP), Generative AI (GPT)
- **Applications:** AI, Health, Climate Risk, Human Mobility, Big Data Analytics, GIScience, Remote Sensing

EXPERIENCE

Consultant, Global Health, World Bank - IFC, Washington, D.C. 05/2024 - Present

Healthcare, Climate, and Tourism Analysis and Modeling

- Processed, analyzed, and mapped large-scale datasets, including healthcare facility data, patient records, climatic disaster data (floods, extreme heat, and droughts), tourism data, wealth index data, population data, and remote sensing images in low- and middle-income countries (LMICs) using **Python, ArcGIS Pro, and Map API**
- [AI] Built predictive models: **Random Forest and Linear Regression** for wealth and population estimation (95%+ accuracy). **XGBoost and Huff Model** for forecasting healthcare facility transactions and market demand. Developed **GPT**-based address-matching pipeline to standardize unstructured data
- Results informed resilience-focused reforms and were adopted by local providers to healthcare service delivery

Research Assistant, Weill Cornell Medical College, Cornell University, New York City 08/2024 - 06/2025

Real-World Evaluation of Cardiovascular Disease Risk Factors

- Collected and processed large datasets on cardiovascular disease, socioeconomic status, and demographic in New York City
- Applied fuzzy string matching to geocode and aggregate patient-level data into neighborhood level for analysis and modeling
- Built **Python** pipelines for scalable, reproducible data processing workflows [GitHub repository]

Research Assistant, GeoAI Lab, University at Buffalo – SUNY, Buffalo 09/2021 - 05/2024

Climatic Disaster Impact Modeling

- Developed data-driven frameworks for assessing spatial and temporal impacts of climatic disasters using human mobility data, remote sensing images, and 311 service request data
- [AI] Applied **time series models (LSTM, Transformer), statistical models (SLM, SEM, GWR), machine learning models (SVM, RF, GRF) with explainable AI (SHAP)** to predict disaster risks and evaluate differential physical and social vulnerabilities of communities during climatic events [GitHub repository]
- Findings supported local agencies in disaster response and first-author paper and paper published and presented at the 2023 and 2025 AAG Conferences

Obesity Prevalence Estimation

- Processed 3GB+ human mobility data and derived diet and physical activity features using **Python**
- [AI] Developed **Geographical Random Forests (GRF) in Python** which takes spatial index into consideration [paper]
- [AI] Implemented **models (OLS, GWR, RF, DNN, GRF)** to predict obesity prevalence with accuracy over 90% through cross validation [GitHub repository]
- First-author paper published and presented at the 2022 GEOMED Conference

Geo-Knowledge-Guided GPT Model

- [AI] Fused geo-knowledge with **GPT models (GPT-2 to GPT-4)** via prompt engineering and question-answering to extract and classify complex location descriptions
- Achieved over 40% improvement in accuracy compared to NER baselines [paper]

Spatial Data Analyst (Intern), MetroDataTech, Shanghai 08/2020 - 09/2020

Educational Facility Assessment

- Analyzed accessibility and land use of educational facilities in Shanghai using GeoPandas in **Python and ArcGIS Pro**
- Generated $30K in business profit by producing actionable insights that informed strategic decisions

Research Assistant, Nanjing Forestry University, Nanjing 09/2018 - 06/2021

Medical Facility Accessibility Assessment

- Processed 2B+ travel records from communities to Wuhan medical facilities during COVID-19 using **Python, SQL, and Map API**
- Assessed the accessibility of medical facilities and developed new service area plans to support decision-making
- First-author paper published and presented at the 2022 UCGIS Symposium